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GENOMICS, TRANSCRIPTOMICS, PROTEOMICS

Delineating the Bacteriome of Packaged and Loose Smokeless Tobacco Products Available in North India

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Abstract

Smokeless tobacco product (STP) consumption is a significant public health threat across the globe. STPs are not only a storehouse of carcinogens and toxicants but also harbor microbes that aid in the conversion of tobacco alkaloids to carcinogenic tobacco-specific nitrosamines (TSNAs), thereby posing a further threat to the health of its consumers. The present study analyzed the bacterial diversity of popular dry and loose STPs by 16S rRNA gene sequencing. This NGS-based investigation revealed four dominant phyla *Actinobacteria, Bacteroidetes, Firmicutes,* and *Proteobacteria* and identified 549 genera, *Prevotella, Bacteroides,* and *Lactobacillus* constituting the core bacteriome of these STPs. The most significantly diverse bacteriome profile was displayed by the loose STP Mainpuri kapoori. The study further predicted the functional attributes of the prevalent genera by Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) algorithm. Genes encoding for nitrate and nitrite reduction and transport enzymes, antibiotic resistance, multi-drug transporters and efflux pumps, secretion of endo- and exotoxin, and other pro-inflammatory molecules were identified. The loose STPs showed the highest level of nitrogen metabolism genes which can contribute to the synthesis of TSNAs. This study reveals the bacteriome of Indian domestic loose STPs that stagger behind in manufacturing and storage stringencies. Our results raise an alarm that the consumption of STPs harboring pathogenic genera can potentially lead to the onset of several oral and systemic diseases. Nevertheless, an in-depth correlation analysis of the microbial diversity of STPs and their elicit impact on consumer health is warranted.

Key points

- *Smokeless tobacco harbors bacteria that aid in synthesis of carcinogenic nitrosamines.*
- *Most diverse bacteriome profile was displayed by loose smokeless tobacco products.*
- *Pathogenic genera in these products can harm the oral and systemic health of users.*

Keywords Microbiome · Smokeless tobacco products · Tobacco-specific nitrosamines · Functional prediction · Oral cancer

Introduction

The consumption of smokeless tobacco (SLT) is one of the foremost reasons for potentially preventable diseases and mortality in South Asian countries as well as globally. The Global Adult Tobacco Survey conducted in 2016–2017 reported that 119.4 million (21.4%) of all adults in India are SLT users, out of which 29.6% are males, while 12.8% are females (GATS-2 2016–17). The Cigarettes and Other Tobacco Products Act (COTPA) adopted in India requires the display of tar and nicotine contents along with the maximum permissible limits on tobacco product packaging (Sect. 7(5)) (Cigarettes and Other Tobacco Products Act 2003). However, no such laws apply to the small-scale industries that manufacture local smokeless tobacco

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products (STPs). Further, this law does not regulate the levels of other toxicants, carcinogens, addictive compounds, microorganisms (bacteria and fungi), and microbial metabolites in tobacco products that have been previously reported in the scientific literature (Kaur et al. 2019). All these components have adverse health outcomes for the users. The microbiological composition of tobacco is noteworthy since microbes naturally colonize this agricultural plant (Chattopadhyay et al. 2021). Moreover, the impact of microbial constituents can be more pronounced in chewable STPs since they are absorbed directly through the oral mucosa and often ingested with the saliva. Hence, the subject requires scientific as well as regulatory attention.

Bacteria generate biomolecules like lipopolysaccharides (LPS), flagellin, endotoxins, and exotoxins such as pneumolysin that are pro-inflammatory and possess carcinogenic potential (Sajid et al. 2021a). Many bacterial species harbor nitrogen metabolism genes that can convert nitrate in the STP to nitrite, a key precursor of tobacco-specific *N*'-nitrosamines (TSNAs) which has been classified by the International Agency for Research on Cancer (IARC) as a group I carcinogen (Song et al. 2016). Moreover, the variations in the climatic and geographic conditions can potentially impact the microbiological composition of the STPs, consequently changing the attributes of the intended constituents of the packaged STPs.

Past investigations of bacterial communities found in STPs primarily employed culture-based methods that could perhaps under-represent the diversity (Mehra et al. 2020; Shetty and Hegde 2015). To overcome this limitation, a culture-independent 16S rDNA metagenomic analysis approach was utilized to analyze the STPs; however, most of the previously published studies focused specifically on STPs from the USA (Han et al. 2016; Rivera et al. 2020; Tyx et al. 2016). Limited scientific literature is available on bacterial contamination in Indian STPs even though the number of SLT consumers has been constantly spiking in the country (Mehra et al. 2020; Monika et al. 2020). The Indian market of STPs is huge with the abundance of various brands and categories thereby prioritizing the need to characterize them for bacterial composition. With this fact, the present study was conducted to perform a comprehensive analysis of the bacterial metagenome of commercially available packaged dry STPs as well as locally procured loose STPs to identify pathogenic or infectious bacterial species and further to perform functional prediction of the genes. Recently, our group published data on bacteriome prevalence in the moist STPs (Sajid et al. 2021b). It is noteworthy here that none of the earlier studies have incorporated locally available loose STPs that lack behind in manufacturing and packaging standards as compared to the packaged STPs making consumers even more susceptible to the toxicological manifestations.

Material and Methods

DNA Extraction of Smokeless Tobacco Products Eight packaged and branded STPs of three different categories, namely Pan masala (PM1-PM3), Gul (G1-G3), and Zarda (Z1-Z2), were procured from retail locations in the metropolitan areas of Delhi, Noida, and Ghaziabad, whereas three varieties of loose tobacco products Mainpuri kapoori (MK), Dohra (D1), and loose-leaf chewing tobacco (T1) were procured from local markets of Mainpuri district, Jaunpur district, and Gautam Budh Nagar district of Uttar Pradesh, respectively. These SLT products are most commonly and widely used in these regions, and the collection was done from December 2019–August 2020. The zip-lock pouches containing loose STPs as well as other unopened packages were stored at 4 °C until further processing. The genomic DNA (gDNA) from different STPs was isolated as per the manufacturer's protocol using DNeasy® PowerSoil® kit (Qiagen, Hilden, Germany). The quantification of DNA was done on a Nanodrop (Thermo, Bangalore, India) and further electrophoresed on an agarose gel before proceeding for PCR amplification.

Amplification of Bacterial 16S rDNA and Construction of Library The amplification of the V_3–V_4 small subunit ribosomal RNA (16S rRNA) hypervariable region of gDNA isolated from all STPs was performed by PCR. The extracted gDNA (40 ng) was amplified using 10 pM of each primer (V13F: 5'-AGAGTT TGATGMTGGCTCAG-3' and V13R: 5'-TTACCGCGGC-MGCSGGCAC-3') and TAQ master mix (high-fidelity DNA polymerase, 0.5 mM dNTPs, 3.2 mM $MgCl_2$, and PCR enzyme buffer). The PCR amplification was performed with initial denaturation at 95 °C for 3 min followed by 25 cycles of denaturation at 95 °C for 15 s, annealing at 60 °C for 15 s, elongation at 72 °C for 2 min, and with a final extension at 72 °C for 10 min. After completion of the reaction, the amplicons were resolved along with DNA markers on an agarose gel and quality was checked using a Nanodrop (Thermo, Bangalore, India).

Next-Generation Sequencing The amplicons from each sample were purified using Ampure beads (Beckman Coulter Inc. Indianapolis, IN) to remove unused primers and Illumina barcoded adapter sequences (Supplemental Table S1) were used to perform additional 8 cycles of PCR to prepare the sequencing libraries. Ampure beads were used to purify the libraries which were quantitated using a Qubit dsDNA High Sensitivity assay kit (Thermo Fisher, Bangalore, India). Sequencing was performed using Illumina Miseq with a 2 × 300PE v3 sequencing kit (Illumina, Inc. San Diego, CA). The sequencing was performed at Biokart India Pvt. Ltd. (Bengaluru, India). BioProject accession number PRJNA787750 was assigned to the raw data that was submitted to the Short Read Archive of NCBI.

Bioinformatic Analysis The raw data were checked for quality using FastQC (version 0.11.9) (Andrews 2010) and MultiQC (version 1.10.1) (Ewels et al. 2016) tools. Trim-Glore (Version 0.6.6) (Krueger 2015) was employed to trim the adapters and low-quality reads. Further processing of these trimmed reads involved chimera removal, merging of paired-end reads, and OTUs (observed taxonomic units) abundance calculation and estimation correction which was accomplished by Quantitative Insights Into Microbial Ecology (QIIME version 1.9.0) (Caporaso et al. 2010) workflows. Highly accurate investigations at the genus level can be conducted following this workflow. The reference database used for the 16S rDNA V_3-V_4 region was NCBI for bacterial identification, and each read is classified based on 97% coverage and identity. The relative abundance of each OTUs is calculated by dividing its value by the total reads per sample. The heatmap, core microbiome, dendrogram, alpha diversity, beta diversity, PCoA plot, rarefaction curve, linear discriminant analysis effect size (LEfSe), random forest, and sparse correlations for compositional data (SparCC) were created and analyzed using the MicrobiomeAnalyst tool (Chong et al. 2020). The pathway analysis was also done using MicrobiomeAnalyst with the PICRUST (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States) option in it (Langille et al. 2013).

Results

Categorization of Smokeless Tobacco Products for Bacteriome Sequence Analysis The core bacteriome of 11 STPs was analyzed using the 16S rDNA amplicon sequencing technique. The sampled STPs included 8 different commercially available brands of dry STPs categorized into Pan masala (PM1, PM2, PM3), Gul (G1, G2, G3), and Zarda (Z1, Z2) groups, whereas the fourth category comprised 3 locally available loose STPs, namely Mainpuri kapoori (MK), Dohra (D1), and loose-leaf chewing tobacco (T1) (Supplemental Table S2).

Alpha and Beta Diversity Analysis of Smokeless Tobacco Products After the pre-processing procedures, the number of amplicons obtained for these STP samples varied between 9,506 and 1,47,437 and has been depicted in an OTU table (Supplemental Fig. S1a, Supplemental Table S3). The alpha diversity is the measure of species variability within a sample, while the difference in composition between the samples is indicated by the beta diversity. The variations in the library sizes across samples were adjusted to enable comparisons of alpha diversity. The data was represented as a rarefaction curve which indicated that the sample MK showed a high number of significant species, whereas PM3 reflected the lowest number of species diversity (Supplemental Fig. S1b). Saturation of species diversity was observed for all the samples examined.

The common alpha diversity indices have been tabulated and graphically represented (Fig. 1a-d, Table 1). The Chao1 index is one of the most comprehensive measures of species richness, and the highest value was recorded for PM1. The Shannon index measures the species abundance and evenness of the samples, and it ranged from 0.86 (Z1) to 4.07 (T1). Similarly, an increase in the homogeneity, as well as species abundance, was also indicated by an increase in the Simpson index of diversity; it ranged between 0.24 (Z1) to 0.96 (G3) for the samples in this study. Interestingly, Gul samples (G1–G3) based on their lower relative moisture content were expected to harbor less diverse populations; however, the results obtained showed that Gul samples exhibited higher overall species diversity than the samples with relatively higher moisture content. This observation can be attributed to the bacteriome contribution of other ingredients of Gul products that are mixed during their manufacturing, such as ash of tendu leaves and sugar or molasses (Singh 2020). However, the four groups of STPs did not display any statistically significant change. The Good's estimator value recorded for all the STPs was > 99% indicating that the majority of bacterial genera harboring a sample have been detected (Table 1).

The beta diversity was measured by comparing the OTU abundance in one sample to another using principal component analysis (PCoA). The samples from the same category of products generally did not cluster together and appeared to form separate groupings (Fig. 2a). The Principal Coordinate Analysis (PCoA) plots were constructed using Permutational Multivariate Analysis of Variance (PERMANOVA) algorithm on Bray–Curtis dissimilarity. The bacteriome of each product category was found to be highly dissimilar (PERMANOVA; $F = 1.716$, $R2 = 0.36387$, $p < 0.083$) (Supplemental Fig. S2). The 3D-PCoA plot exhibited close association between samples of Gul and loose tobacco category and therefore, restrain more related bacteriome profiles. Two samples from Pan masala (PM1 and PM2) were dominated by *Prevotella* and they clustered separately from the third sample (PM3) which showed the prevalence of *Ruminococcus*. The two samples of Zarda were significantly distinct from each other, *Acinetobacter* was dominant in Z1, while *Prevotella* was abundant in Z2. Next, one sample from Gul (G3) and one sample from loose tobacco (T1) were also found to be close to each other and clustered separately from the other samples (Fig. 2a).

A decision tree was constructed by employing the 'random forest' algorithm that will enable us to validate the similarities and dissimilarities in bacteriome among the four categories of STPs (Fig. 2b). In the error plot the red line indicates the overall genera present in the analyzed STPs,

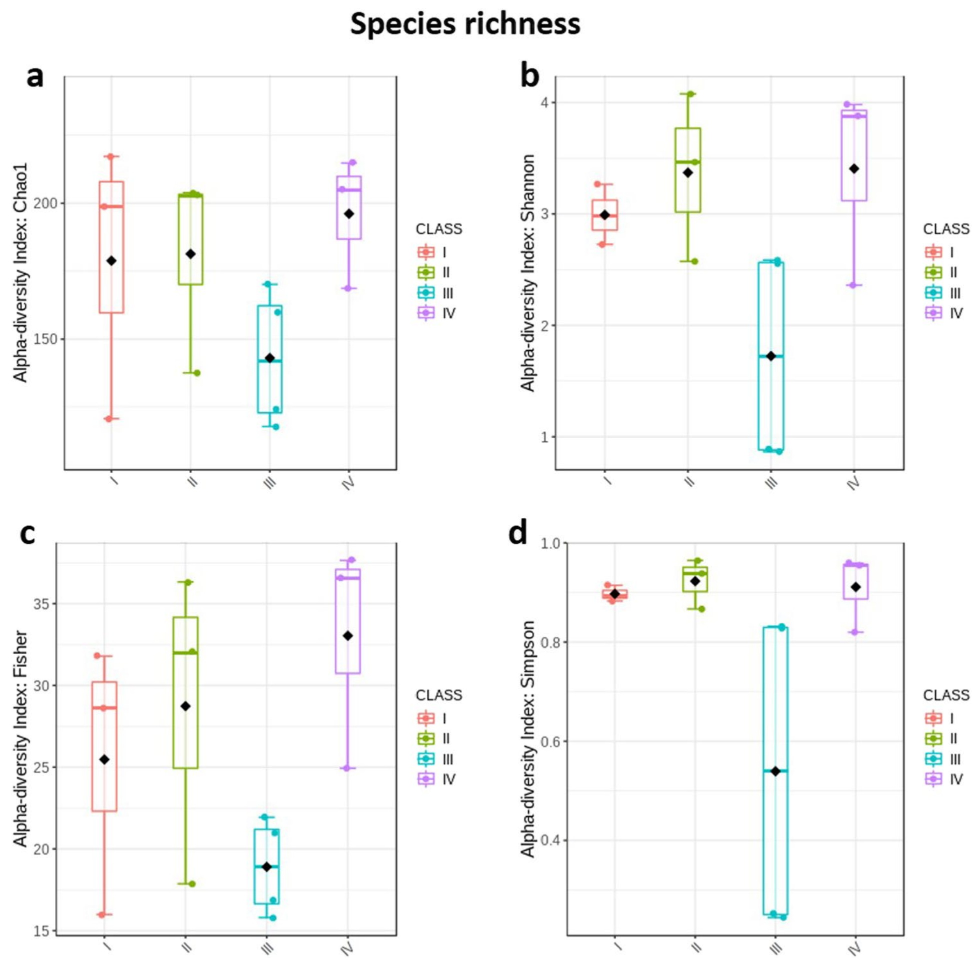


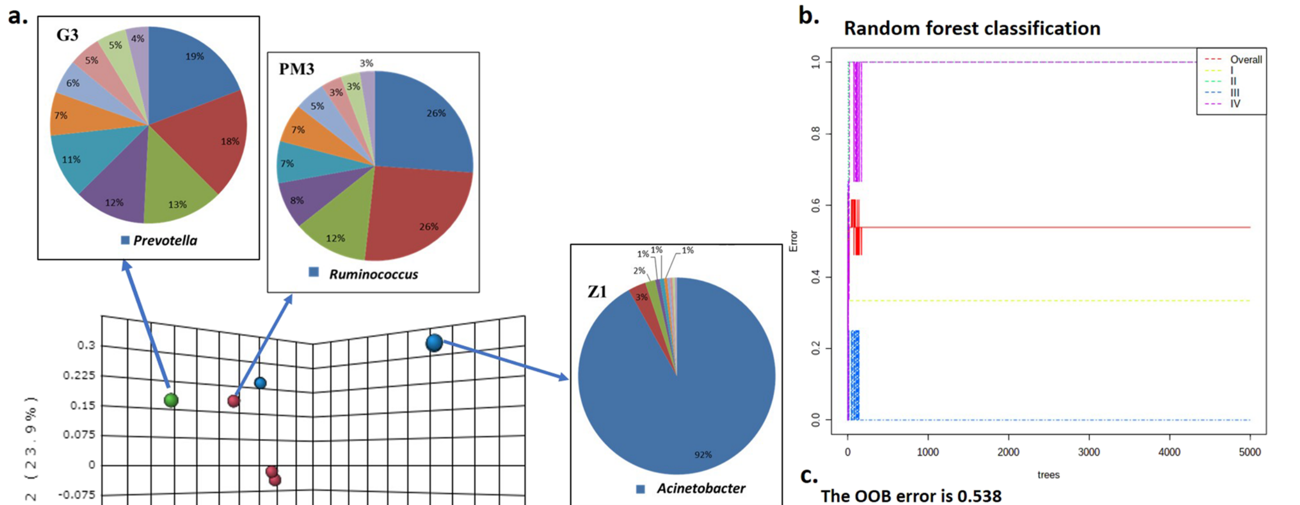
Fig. 1 ***Bacteriome α-diversity (within-sample) estimation using 16S rRNA sequence analysis in smokeless tobacco products.*** (**a**-**d**) Metrics for estimating α-diversity. Error bars represent standard deviation

Table 1 Diversity indices and estimators among dry smokeless tobacco products

S.No	Category	Sample ID	Diversity indices and estimators				
			Chao1	Fisher	Shannon	Simpson	Good's coverage (%)
1	Pan masala	PM1	217.2308	28.6300	2.9814	0.8942	99.6068
		PM2	198.6000	31.7891	3.2657	0.9144	99.6493
		PM3	120.7500	15.9892	2.7269	0.8829	99.7343
2	Gul	G1	137.5882	17.8672	2.5743	0.8666	99.6812
		G2	202.6471	32.0021	3.4603	0.9370	99.7131
		G3	203.8667	36.3252	4.0771	0.9646	99.9150
3	Zarda	Z1	159.7143	20.9507	0.8656	0.2444	99.6068
		Z2	117.8667	16.9233	2.5855	0.8318	99.7556
4	Loose tobacco	MK	214.8000	37.6448	3.8777	0.9539	99.8193
		D1	168.6667	24.9326	2.3599	0.8198	99.6599
		T1	204.9565	36.5444	3.9815	0.9539	99.8937

whereas the yellow, green, blue and magenta lines represent the distinct genera present in Pan masala, Gul, Zarda and loose tobacco, respectively.

Bacterial Community Composition of Smokeless Tobacco Products The metagenomic analysis of STPs showed an extensive array of taxonomic diversity. A total of 32 phyla were identified



		I	II	III	IV	Class error
Pan masala	I	2	0	1.0	0.0	0.333
Gul	II	0	0	1.0	2.0	1.0
Zarda	III	0	0	4.0	0.0	0.0
Loose tobacco	IV	1	2	0.0	0.0	1.0

Fig. 2 ***β-diversity (between different samples) estimation in smokeless tobacco products*** **a**) 3-D Principal Coordinate Analysis (PCoA) plots showing the similarity relations among the bacteriome composition of the four categories of STPs generated using unweighted UniFrac distances of 16S rRNA genes using from the phylogenetic tree. The samples of each group are represented by different colors as indicated in the above figure. Three pie charts, one from Pan masala sample (PM3), one Gul (G3), and one Zarda sample (Z1), are shown at the level of genera. (**b**) The error plot generated from random forest simulation results. The number of trees run was plotted on X axis, whereas Y axis represents the corresponding error. Overall genera present in STPs was represented by a red line, yellow line indicates the distinct genera present in Pan masala, green line showed specific genera of Gul products, blue-line represent unique genera of Zarda samples and magenta line specify exclusive genera of loose STPs. (**c**) Out-of-bag (OOB) score for validation of random forest model

in 11 STPs; however, OTUs > 10 reads are represented in Fig. 3a, showing the relative abundance of 21 phyla in the STPs. Four major phyla were reported in all STPs, namely *Actinobacteria, Bacteroidetes, Firmicutes,* and *Proteobacteria*. The other prominent phyla were *Acidobacteria, Chloroflexi, Planctomycetes, Cyanobacteria, Fusobacteria, Gemmatimonadetes, Sprirochaetes, Nitrospirae,* and *Verrucomicrobia*.

The phyla *Firmicutes* was significantly abundant in G1 (65%), PM3 (58%), D1 (45%), Z2 (43%), PM1 (43%), and PM2 (41%). Further, MK (31%), G2 (29%), G3 (18%), and T1 (15%) were shown to harbor a considerable fraction of bacterial species belonging to *Firmicutes* phyla. The *Proteobacteria* phyla was predominant in Z1 (88%), D1 (47%), T1 (44%), G3 (43%) G2 (37%), and MK (37%). The prevalence of *Bacteroidetes* phyla was found to be elevated in Z2 (41%); however, its relative abundance decreased to 4% in both Z1 and D1 samples. This phylum was significantly present in the products of Pan masala category (PM1- 28%; PM2- 28%; PM3- 28%), Gul category (G1- 21%; G2- 19%; G3- 19%), and in loose-leaf chewing tobacco (T1-19%). The phyla *Actinobacteria* was identified in all STPs having a prevalence in the range of 1–10%.

A total of 61 classes were identified across all STPs out of which 26 classes with OTUs > 10 reads have been graphically represented in Supplemental Fig. S3. Furthermore, 261 bacterial families were identified and OTUs > 50 in all STPs were represented in their relative abundance. The *γ–Proteobacteria* accounted for the predominant class in all STPS showing the highest abundance in Z1 (82%) and D1 (40%) samples. The other prevalent classes identified were *Bacilli, Clostridia, Bacteroidia* and *α–Proteobacteria*. The Pan masala products PM1 and PM2 showed an increased abundance of family *Moraxellaceae* of the *γ–Proteobacteria* class (21% and 20% respectively), whereas PM3 showed an abundance of *Ruminococcaceae* (27%) of class *Clostridia*. The Gul samples G1 and G2 were observed to be dominated by the *Lactobacillaceae* family. Further, loose-leaf chewing tobacco (T1) showed a very similar bacterial community profile with G3 at the family level and were dominated by *Rhodobacteriaceae, Moraxellaceae* and *Prevotellaceae*. One of the local products MK contained a high percentage (29%) of *Enterobacteriaceae* of the class *γ–Proteobacteria* and D1 showed an abundance of *Bacillaceacae* (19%) of the class *Bacilli* (Supplemental Fig. S4).

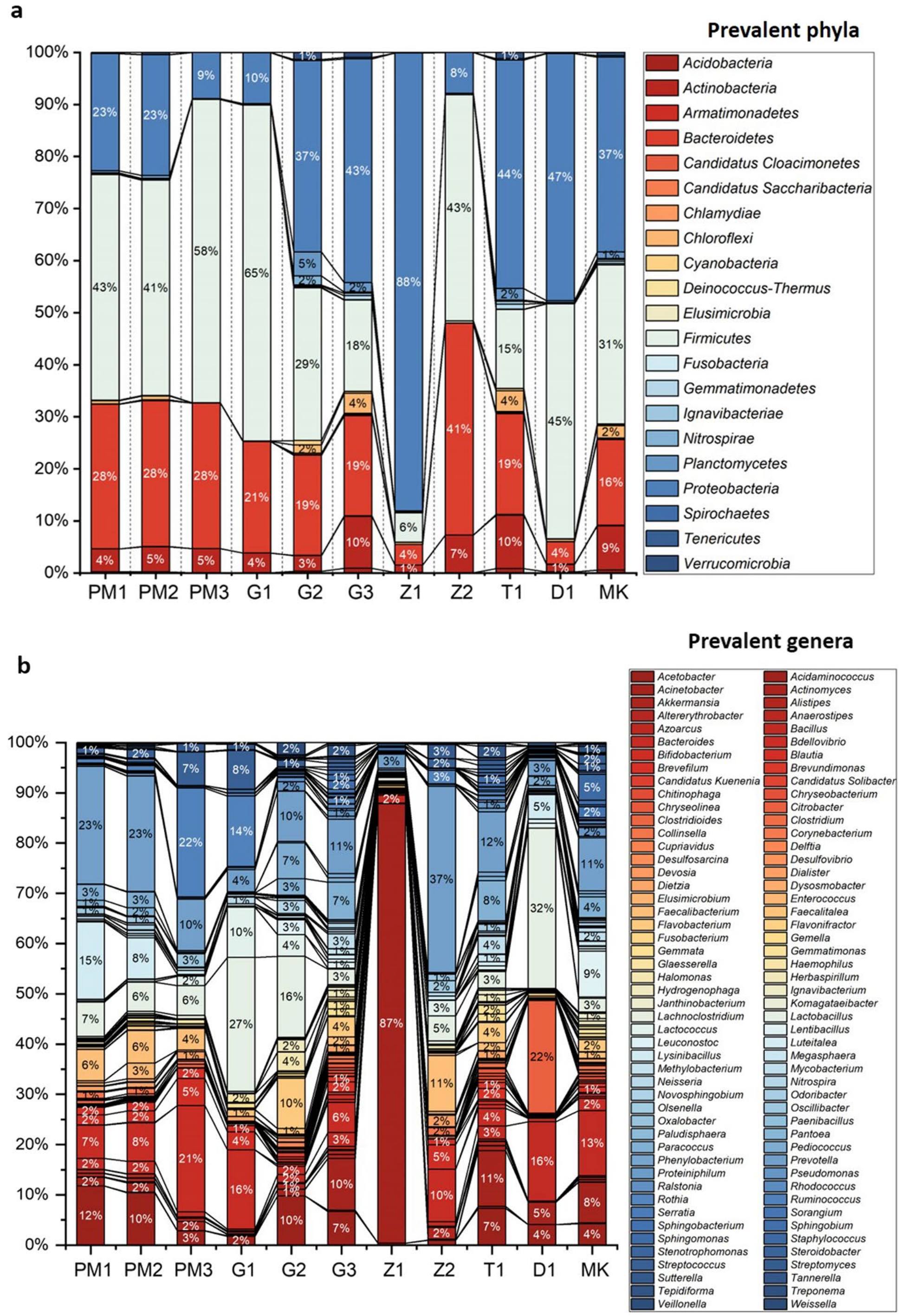


Fig. 3 ***Bacterial phyla and genera identified in smokeless tobacco products.*** The stacked bar showed the relative abundance of (**a**) bacterial phyla (OTUs > 10 reads) and (**b**) bacterial genera (OTUs > 100 reads) identified in each STP. Each bacterial phyla and genera has been symbolized by a different color in the stacked bar graph with the connecting lines. The entire relative abundance was calculated as 100% for each product

In the bacteriome of the 11 STPs, a total of 549 genera were identified, out of which 98 genera are presented in Fig. 3b on the basis of their relative abundance (OTUs > 100 reads). The genus *Acinetobacter* was found to be significantly high in Z1 (87%). The genus *Bacteroides* was abundant in PM3 (21%), G1 (16%) and Z2 (10%). Significant levels of *Prevotella* were observed in Z2 (37%), PM1 (23%) and PM2 (23%). The level of genera *Lactobacillus* was higher in D1 (32%) and G1 (27%). The other important genera observed in these STPs were *Ruminococcus* (PM3-22%, G1-14%), *Bacillus* (D-16%, MK-13%), *Faecalibacterium* (Z2-11%), *Citrobacter* (D1-22%) and *Acetobacter* (PM1-12%, PM2-10%, G2-10%).

Core Bacteriome of Smokeless Tobacco Products Although the STPs varied in their physiochemical characteristics and metagenomic composition, a core bacteriome was identified in the dry as well as loose STPs that remained unaltered across the different groups sampled in this study. Twenty core bacterial genera were identified as the core bacteriome of these STPs having sample prevalence (≥ 20%) and relative abundance (≥ 0.2%). The prevalence of *Prevotella, Bacteroides,* and *Lactobacillus* was observed to be abundant in these STPs. Moreover, *Acetobacter, Faecalibacterium, Bifidobacterium, Blautia, Acinetobacter*, *Streptococcus,* and *Oscillibacter* were also found to be the dominant core bacterial genera in the analyzed STPs (Fig. 4).

Clustering Analysis of Bacterial Genera Associated with Smokeless Tobacco Products The 11 STPs were clustered into three groups, represented by a dendrogram on the basis of the Bray–Curtis index to determine the most relevant correlations at the OTU level amongst the STPs analyzed (Supplemental Fig. S5). Group-I consisted of three Pan masala products PM1, PM2, and PM3; however, PM3 clustered into a subgroup with one of the Gul products G1. The Zarda samples Z2 and Z_2 dup also clustered in another subgroup of Group-I. The other two Gul products clustered together with loose tobacco products in Group-II. D1 and G2 showed similar bacterial composition while T1, MK, G3 clustered separately in another subgroup. Group-III comprised of the Zarda samples Z1 and Z_1 dup which validated the reproducibility of the sequencing data.

Moreover, the differentially abundant bacterial genera across the different STPs were depicted using a hierarchical clustering heatmap (Supplemental Fig. S6). The dominant genera in Pan masala samples included *Lysinibacillus, Anaerostipes, Clostridioides, Tannerela, Ruminococcus, Novosphingobium, Oscillibacter,* and *Bacteriodes*. The Gul samples showed an abundance of genera such as *Lactococcus, Gemella, Haemophilus, Streptococcus, Pediococcus, Weissella, Halomonas, Leuconostoc, Flavobacterium,* and *Hydrogenophaga.* The Zarda products demonstrated the dominance of *Acinetobacter, Sutterella, Alistipes*, *Dialister, Faecalibacterium,* and *Prevotella*. Each loose tobacco product exhibited diverse bacterial genera. D1 showed abundance of genera *Bacillus* and *Lactobacillus* whereas, MK was dominated by the genera *Sphingobacterium, Staphylococcus, Streptomyces, Fusobacterium, Neisseria, Bacillus, and Pseudomonas*. T1 was majorly constituted by the genera *Megasphaera, Mycobacterium, Tepidiforma, Clostridium,* and *Paracoccus*.

The genera having clinical relevance like *Staphylococcus*, *Fusobacterium,* and *Pseudomonas* were abundant in MK, *whereas Haemophilus* and *Streptococcus* were predominant in G3.

Co-occurrence Network Analysis at Genera Level Potential interactions within the bacterial communities present in the STPs were identified by constructing a co-occurrence network at the level of genus. For this purpose a robust compositional method SparCC correlation coefficient (r) was used

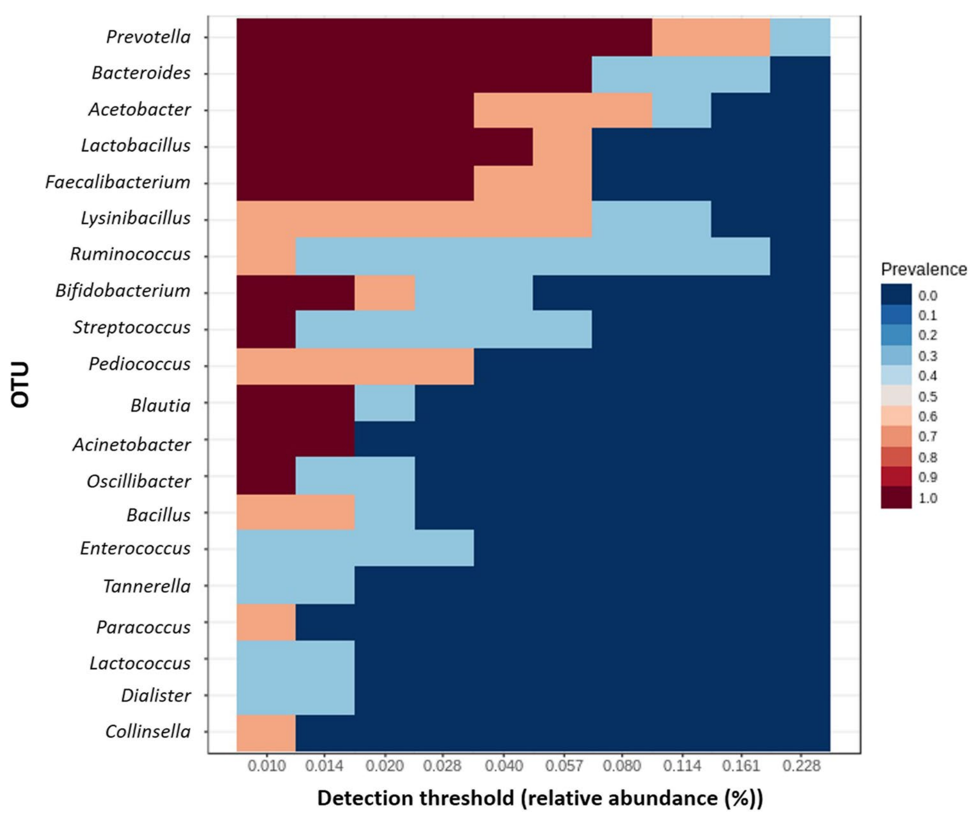


Fig. 4 ***Core bacteriome of smokeless tobacco products.*** The core bacterial genera in STPs was estimated using the parameters of sample prevalence (≥ 20%) and relative abundance (≥ 0.2%). Heatmap illustrating the detection threshold and relative abundances of the most dominant bacterial genera in the analyzed STPs. The color key shows the range of threshold relative abundance of the individual values

to create a sparse correlation network (Friedman and Alm 2012). It was observed that the prevalence of 224 genera were significantly different between the 4 groups of STPs (Supplemental Fig. S7a). Supplemental Table S3 depicts 133 positive and 91 negative significant correlations (=*r* |> 0.3 and *p*-value < 0.05) between 224 genera. The genus *Acinetobacter* was found to be correlated positively with *Sphingomonas* (r = 0.9837, p = 0.0099), *Streptomyces* (r = 0.8441, p = 0.0198), *Baekduia* (r = 0.7739, p = 0.0099), *Xanthomonas* (r = 0.7166, p = 0.0099), *Herbaspirillum* (r = 0.7, p = 0.0099), and *Rhizobium* (r = 0.5674, p = 0.0396) while it was negatively correlated with *Novosphingobium* (r = -0.6889, p = 0.0099) and *Pontibacter* (r = -0.6581, p = 0.0297). *Prevotella* showed positive correlation with *Veillonella* (r = 0.9378, p = 0.0099) and *Neisseria* (r = 0.9106, p = 0.0198), whereas *Streptococcus* was positively associated with *Haemophilus* (r = 0.9809, p = 0.0099), *Gemella* (r = 0.9804, p = 0.0099), *Bacteroides* (r = 0.9745, p = 0.0099) and *Clostridium* (r = 0.9226, p = 0.0099). *Bifidobacterium* displayed positive concurrence with *Dialister* (r = 0.9906, p = 0.0099), *Bacteroides* (r = 0.9958, p = 0.0099), *Clostridium* (r = 0.943, p = 0.0099), *Fusobacterium* (r = 0.9112, p = 0.0099) and *Pseudomonas* (r = 0.8346, p = 0.0099) (Fig. 5). The correlation network plot is interactive and upon selection of the genera, *Acinetobacter,* its highest abundance was observed in Zarda products followed by Gul, loose tobacco and the least presence was seen in the Pan masala products (Supplemental Fig. S7b).

Biomarker Discovery The nonparametric statistical method linear discriminant analysis effect size (LEfSe) was employed to identify the robust biomarkers of STPs. Figure 6a represents significant taxa in these STPs identified by using the cutoff values: FDR-adjusted *p*-value < 0.1 and linear discriminant analysis (LDA) > 4.0. The Pan masala products displayed highest LDA score for *Acinetobacter* and *Lysinibacillus* whereas loose tobacco samples had high LDA scores for *Bacillus* and *Citrobacter*. Further, the LDA scores of *Pseudomonas* and *Gemmatimonas* were also dominant in loose tobacco products. *Pediococcus, Weissella,* and *Leuconostoc* exhibited dominance in Gul products.

Thereafter, a random forest algorithm was applied to identify bacterial genera that differentiate between phenotypes. A pattern of changes in the significant genera was seen across different groups of STPs with random forest model, and the genera *Acetobacter, Weissella,* and *Candidatus_Solibacter* were identified because of their higher predictive values as the most decisive discriminated genera (Fig. 6b).

Functional Potential of Bacteriome Associated with Smokeless Tobacco Products PICRUSt assists in inferring the metabolic potential of bacteriome from 16S rDNA sequencing data based on sequence similarity of the identified microbes with microbes whose whole genome has been sequenced and is based on Greengenes annotated OTUs (Langille et al. 2013). The

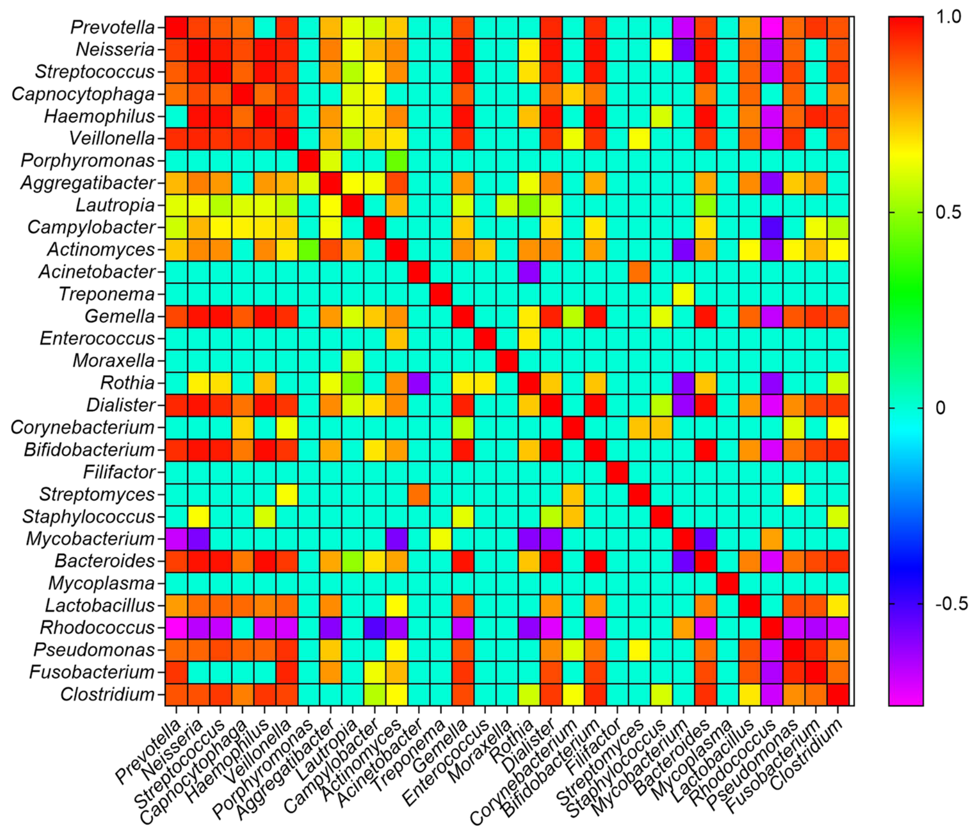


Fig. 5 ***Correlation matrix analysis.*** The correlation matrix was plotted using the SparCC correlation coefficient to analyze any potential interaction between any two genera. All diagonal elements equal to + 1. The heatmap to the right of the plot designates whether the correlation between the genera were high (red) or low (blue)

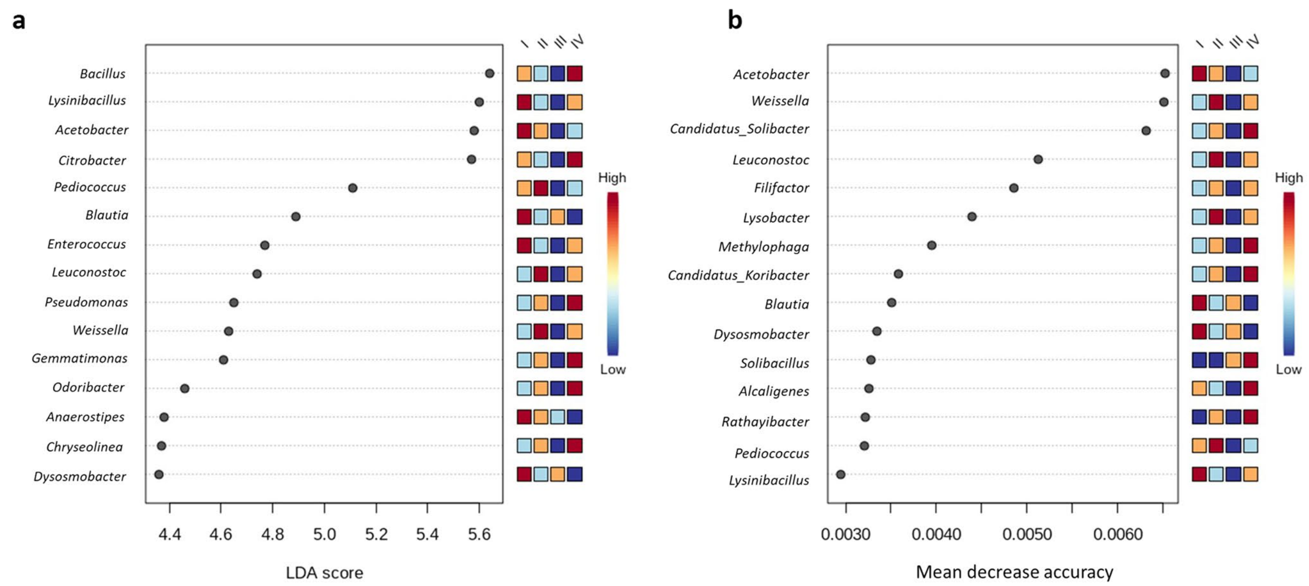


Fig. 6 ***Biomarker analysis of dry smokeless tobacco products-linked bacteriome***. (**a**) Linear discriminant analysis effect size (LEfSe) of bacteriome present in the analyzed STPs. The significant 15 genera were ranked in declining order as per their LDA scores (x-axis). (LEfSe parameters; Taxonomic level-genus, FDR-adjusted *p*-value cut off <0.1, log LDA score >4.0). (**b**) The significant feature was identified by random forest analysis. The variable importance calculated by mean decrease in accuracy of predictor genera in the random forest model and top 15 genera were ranked in increasing order as per their mean decrease accuracy value (x-axis). The heatmap to the right of the plot designates whether the genera were higher (red) or lower (blue) in each group of STP

MicrobiomeAnalyst computed the metagenome contributions for all dry and loose STPs based on KEGG (Kyoto Encyclopedia of Genes and Genomes) orthology (KO term) (Kanehisa and Goto 2000). The result containing KO abundance level was generated and a total of 3785 KO terms were identified in the imputed metagenome of dry STPs and loose tobacco products leading to identification of several KEGG metabolic pathways (Supplemental Table S3). MicrobiomeAnalyst tool aided in monitoring the relative abundance of these metabolic pathways in different STPs (Supplemental Fig. S8).

Nitrogen Metabolism Potential of STP-Associated Bacteriome The nitrogen metabolism pathway plays the most important role in the formation of TSNAs. The loose STPs, MK and T1 showed abundance of the dissimilatory nitrate reduction pathway genes (*narZ*, *narJ,* and *narI*) whereas, PM2, PM3 and Z2 products showed a significant number of imputed genes of the nitrate reduction pathway (Fig. 7). The assimilatory nitrate reductases genes *nasA* and *nasB* were also predicted by PICRUSt and abundantly monitored in Pan masala (PM1, PM2, PM3) category and in one Zarda product Z2. The predicted genes of nitrite reductases including *nirA*, *nirB*, *nirD*, *nirK,* and *nrfA* were found to be abundant in all three Pan masala products whereas, *nirB*, *nirD,* and *nrfA* were prevalent in one Zarda (Z2) product. The genes *nirA and nirK* were prevalent in loose tobacco products (MK and T1). Denitrification is an important step in conversion of nitrogenous compounds (nitrate, nitrite, and ammonia) to nitrogen gas (N_2). All STPs of the Pan masala category and one Zarda category (Z2) contained imputed genes related to denitrification *nosZ* (K00376), *norB* (K04561), and *norC* (K02305) (Supplemental Table S3). Further, all loose tobacco products were abundant in nitrogen fixation related genes *nifD1* (K02586), *nifH* (K02588), and *nifK* (K02591). The *nifD1* and *nifK* genes were prevalent in MK and T, whereas *nifH* was prevalent in D1 (Fig. 7).

Antibiotic Drug Resistance Abundance in STP-Associated Bacteriome All three products of the Pan masala category, one Zarda product (Z2) and one loose tobacco product (MK) displayed the highest number of antibiotic resistance genes (Fig. 8). The most prolific antibiotic resistance genes were penicillin-binding protein 1A (*mrcA*) (K05366), serine-type D-Ala-D-Ala carboxypeptidase/endopeptidase (penicillin-binding protein 4) (*dacB*) (K07259), multiple antibiotic resistance protein (*marC*) (K05595), beta-lactamase class C (*ampC*) (K01467), and alanine racemase (*alr*) (K01775) (Supplemental Table S3). The product PM2 showed genes such as MFS transporter, enterobactin (siderophore) exporter (*entS*, K08225), macrolide phosphotransferase (*mph*, K06979), and macrolide efflux protein (*ykuC*, K08217) which had higher predicted prevalence. The product MK exhibited the prevalence of MFS transporter multidrug resistance protein (*bcr*, *tcaB*, K07552), MFS transporter, beta-lactamase induction signal transducer (*ampG*, K08218), stage V sporulation protein D (*spoVD*, K08384), and zinc D-Ala-D-Ala dipeptidase (*vanX*, K08641) (Fig. 8).

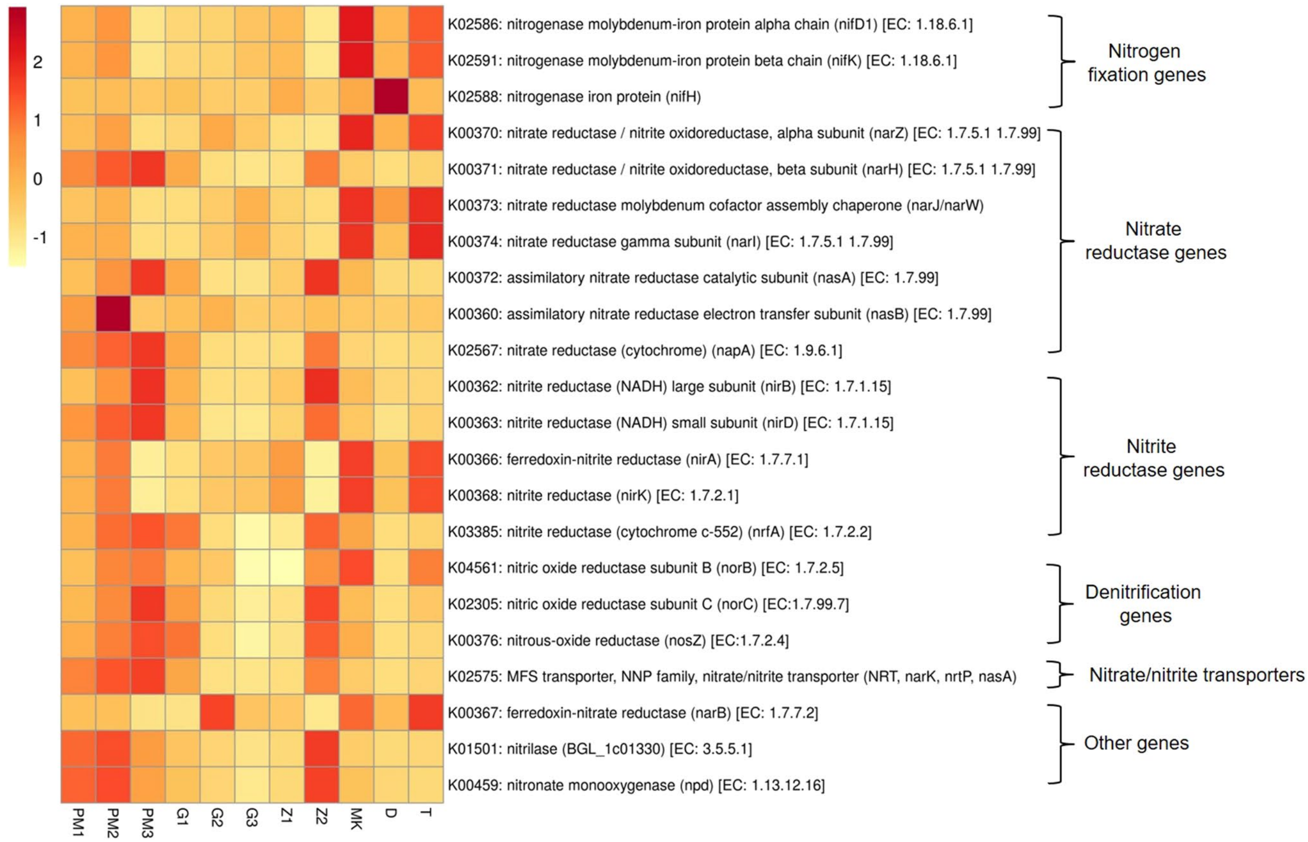


Fig. 7 ***Relative abundance of nitrogen metabolism pathway genes in smokeless tobacco products.*** Heatmap displaying the predicted genes identified using PICRUSt (y-axis) based on KEGG database in each sample of the STP (x-axis). Ward's clustering algorithm and Euclidean distance measure were used to generate the hierarchical tree. Each column represents a SLT sample and each row nitrogen metabolism gene with relative abundance indicated by color bar

The sample Z2 displayed the presence of several vancomycin resistance genes, penicillin resistance genes and β-lactam resistance genes like penicillin-binding proteins (*mrdA*, KO5515; *mrcA*, KO5366), penicillin G amidase (*pga-2*, K01434), vancomycin resistance-associated response regulator (*vraR*, K07694), and beta-lactamase class C (*ampC*, K01467). The product D1 displayed multiple antibiotic resistance proteins (*marC*, K05595). The Gul product G1 showed a dominance of gene coding for MFS transporter, multidrug resistance protein (*norB*, K08170), whereas the gene zinc D-Ala-D-Ala carboxypeptidase (*vanY*, K07260) was observed to be prevalent in G2 (Fig. 8).

Toxic and Pro-inflammatory Effects of Smokeless Tobacco Products-Associated Bacteriome

Lipopolysaccharide (LPS), a Gram-negative bacterial cell wall component, can induce inflammation by mediating the production of pro-inflammatory cytokines in several cancer types (De Waal et al. 2020; Melkamu et al. 2013). The LPS inflammatory activity is attributable to the lipid-A component. The STPs such as PM2, G2, Z2, and MK showed a high abundance of the ABC transporter complex (*lptBFG*) which aids the transportation of LPS from the inner to the outer membrane. The functional prediction also revealed that the abundance of genes like lipid-A-disaccharide synthase (*lpxB*) and Kdo2-lipid IVA lauroyltransferase/acyltransferase (*lpxL*) which are involved in LPS biosynthesis to be significantly increased in Pan masala (PM1, PM2, PM3) and Z2 products (Supplemental Fig. S9).

The flagellin coding gene *fliC*, which assists in the flagellar assembly for the motility of bacteria, was seen to be abundant in Pan masala and Z2 (Supplemental Fig. S9, Supplemental Table S3). Genes participating in the synthesis of lipoteichoic acid (K03739, K03740), peptidoglycan, and teichoic acid were not observed in any of the analyzed STPs (Supplemental Table S3).

Furthermore, the prevalence of another important category of genes related to bacterial toxins was analyzed since they can cause inflammation and induce genomic damage in epithelial cells leading to neoplastic transformations (La Rosa et al. 2020). The dry STPs Pan masala (PM1, PM2, PM3), G1, and Z2 showed a high abundance of toxin genes

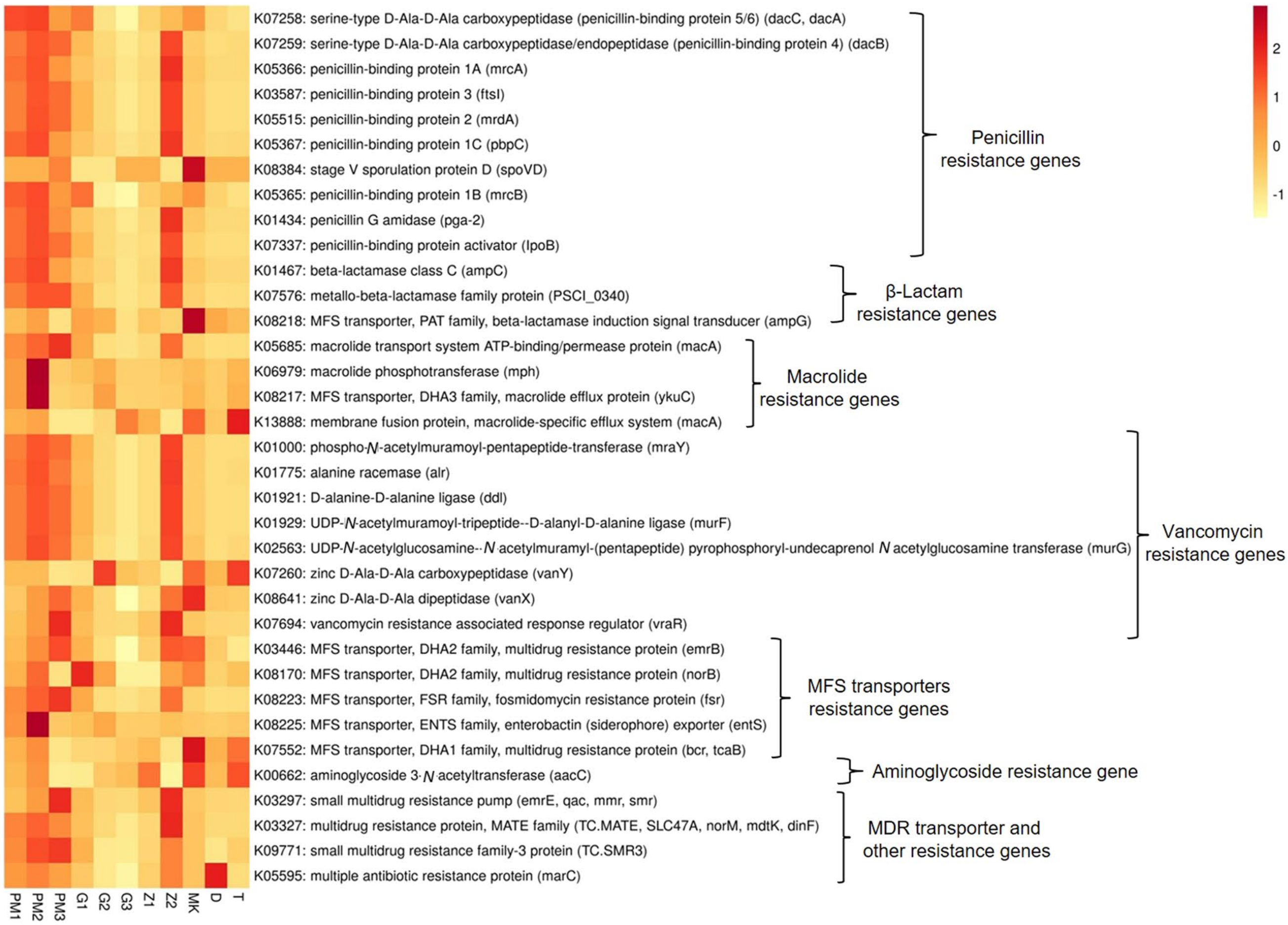


Fig. 8 ***Relative abundance of antibiotics resistance genes in smokeless tobacco products.*** Heatmap displaying the predicted genes identified using PICRUSt (y-axis) based on KEGG database in each sample of the STP (x-axis). Ward's clustering algorithm and Euclidean distance measure were used to generate the hierarchical tree. Each column represents a SLT sample and each row antibiotic resistance gene with relative abundance indicated by color bar

like α-hemolysin/cyclolysin transporter gene *tolC* (K12340), and neuraminidase1 gene (*NEU1*, K01186) (Supplemental Fig. S9).

Association of STP-Associated Bacteriome with Human-Intrinsic Factors

Next, a Taxon Set Enrichment analysis was performed using the MicrobiomeAnalyst tool, where all significant genera found in these STPs were examined across 239 taxa set related to host-intrinsic factors like diseases, and 53 taxa set associated with microbiome-intrinsic factors such as microbe motility and shape (Chong et al. 2020). The microbiome profile of colorectal carcinogenesis, Type I diabetes, and chronic obstructive pulmonary disease (COPD) were observed to be correlated with the bacteriome of these STPs, however, the association was not significant (Table 2). Further, a significant correlation was observed in the STP bacteriome with that of indole producers and mucin degraders (Table 2).

Discussion

The present study was conducted to characterize the bacterial populations that harbor various types of STPs and to predict opportunistic or pathogenic bacterial genera that could lead to chronic inflammation and subsequent induction of cancer. High-throughput sequencing of the 16S rDNA (V_3-V_4 segment) region was conducted on the bacterial genomic DNA isolated from STPs to determine their bacteriome. This study is the first instance of a comprehensive community profiling, comparative analysis, and functional prediction conducted on the bacteriome associated with dry STPs from the Indian market by using the MicrobiomeAnalyst tool (Dhariwal et al. 2017).

Table 2 Correlation of host-intrinsic factors (diseases) and microbiome-intrinsic factors with bacteriome of smokeless tobacco products

Correlation with bacteriome of STPs	Total	Expected	Hits	Raw *p*	Holm *p*	FDR
Host-intrinsic factors (diseases)						
Colorectal carcinogenesis (increase)	7	1.29	6	0.000218	0.052	0.044
Liver cirrhosis (China, decrease)	52	9.56	20	0.00043	0.102	0.044
Type I diabetes (increase)	13	2.39	8	0.00065	0.153	0.044
Chronic obstructive pulmonary disease (COPD)	8	1.47	6	0.000739	0.174	0.044
Alzheimer's (increase)	6	1.1	5	0.00104	0.243	0.0494
Crohn's disease (increase)	21	3.86	10	0.00196	0.458	0.0779
Microbiome-intrinsic factors						
Indole producers	83	12.6	29	3.71E-06	0.000193	0.000193
Mucin degraders	11	1.67	7	0.000326	0.0166	0.00848
Folate consumers	1	0.152	1	0.152	1	1
Riboflavin producers	5	0.758	1	0.516	1	1
Butyrate producers	14	2.12	2	0.65	1	1

In this analysis, we included the STPs which are dry and have weakly basic pH, grouped into three categories Pan masala, Gul, and Zarda. The fourth group comprised loosely available STPs procured from local markets. Pan masala is a mixture of areca nut, catechu, lime, spices, and flavoring agents available in the form of a dry powder. Even though it is labelled to be nicotine free in composition, several testing laboratories have reported otherwise (Mukherjea et al. 2015; Rumi 2019). The second group is comprised of products from the Gul category, which is a tobacco-containing dentifrice used not only for teeth cleaning but also to alleviate acute dental pain owing to its nicotine-induced analgesic effect. Characterizing microbial composition of Gul products is crucial since it is applied multiple times daily onto the affected teeth and adjacent gum/mucosa and has also shown a statistically significant positive association with the development of oral potentially malignant disorders (Nethan et al. 2018). Further, the third group is comprised of products from the Zarda category which is partially fermented tobacco produced by boiling tobacco leaves in water with spices and lime until evaporation. This category has been previously reported to contain high levels of TSNAs and nicotine, therefore deciphering its microbial profile will be intriguing (Nasrin et al. 2020; Tricker and Preussmann 1988). It is also often consumed in combination with Pan masala, thereby enhancing the exposure levels of TSNAs to the consumers (Stepanov et al. 2005). The products in the fourth group are noteworthy since they are loosely available in local markets and their manufacturing and marketing practices are not regulated, unlike the products in the other three categories. Their moisture content is comparatively higher than other sampled STPs, which can play a crucial factor in facilitating the growth of microorganisms (Han et al. 2016). Moreover, these products have not been analyzed in any previous study.

The alpha diversity was established and the species richness was observed to be higher in loose STPs such as Mainpuri kapoori (MK) and loose-leaf chewing tobacco (T1) that have high moisture content compared to products having low moisture levels like Pan masala and Zarda. *Actinobacteria, Firmicutes, Proteobacteria,* and *Bacteroidetes* were the most abundant phyla identified in the STPs analyzed in the present study. The predominance of these four phyla has also been observed in earlier studies conducted on dry snuff products procured from the USA and North India. The selection of dry products in the present study was based on previous findings that dry snuff products have greater diversity indices at the family level than moist products (Tyx et al. 2016). The most prevalent bacterial families in the analyzed STPs included *Bacillaceae, Lactobacillaceae, Enterococcaceae,* and *Moraxellaceae*. These were also identified in the metagenomic composition of STPs procured from the USA and North India, but the relative abundance was variable (Mehra et al. 2020; Tyx et al. 2016).

The bacterial genera characterized in this study, including *Acinetobacter, Bacteroides, Prevotella, Lactobacillus, Ruminococcus, Bacillus, Faecalibacterium, Citrobacter,* and *Acetobacter* have also been reported in previous studies that analyzed STPs. A study on American moist-snuff products metagenomic analysis observed several predominant genera *Tetragenococcus, Aerococcus, Alloiococcus*, and *Staphylococcus* (Tyx et al. 2016), while another metagenomic study on American moist-snuff products identified genera *Paenibacillus*, *Oceanobacillus*, and *Bacillus* (Al-Hebshi et al. 2017). Further, a study on Indian STPs (chewable tobacco, snus, and snuff) established the abundance of genera *Staphylococcus, Bacillus, Corynebacterium, Virgibacillus, Brevibacterium, Rothia, Veillonella*, and *Fusobacterium* (Monika et al. 2020). The differences can be attributed to the approach since the V_3-V_4 hyper-variable region belonging to the 16S rRNA gene segment was sequenced in the current study, whereas previous studies characterized the SLT microbiome by selecting other regions of the 16S rRNA gene (Al-Hebshi et al. 2017; Monika et al. 2020; Tyx et al. 2016). Although the present study has identified the bacterial population only till the taxonomic classification of genus, it is alarming that several species belonging to the genera identified in the current analysis of STPs have been reported to be pathogenic. *Bacillus licheniformis* has been reported to be pathogenic in immunosuppressed mice, and accidental exposure can establish infections in susceptible individuals (Agerholm et al. 1997). *Staphylococcus epidermidis* and *Staphylococcus hominis* strains were reported to be involved in the formation of carcinogenic TSNAs by reducing nitrates to nitrites (Han et al. 2016). Organisms such as *Streptococcus mutans* and *Actinomyces* spp are the primary colonizers in oral bacterial infections including dental caries, gingivitis (inflammation of gums), periodontitis, and dental abscess. *Fusobacterium nucleatum* and *Prevotella intermedia* are the secondary colonizers that eventually lead to the formation of biofilm and plaque (Reynolds-Campbell et al. 2017). Genera such as *Prevotella* and *Streptococcus* were highly abundant in all the STPs and comprise their core bacteriome. Further, β- diversity analysis of STPs revealed that the samples from the same product category generally did not cluster together and appeared to form separate groupings depicting distinct bacterial populations. This variability can be credited to the fact the samples of the same product category belonged to different brands/manufacturers and hence vary in their composition.

Several significant relationships within the bacteriome of STPs were identified on the basis of the genera co-occurrence network analysis. Positive correlation was shown in the genera involved in the development of oral diseases independent of their phyla. In contrast with previously published scientific literature relevant to the bacteriome of STPs, this study primarily identified 25 distinguishing genera associated directly with the STPs analyzed in the current study, by using the LEfSe method and random forest analysis.

The next section of the study focuses on the predictive functional potential of the bacteriome identified in the STPs. Certain microorganisms found in tobacco are known to contribute to the formation of TSNAs, endotoxins, and other pro-inflammatory molecules (Vishwakarma and Verma 2021). The predictions made in this study will aid in the understanding of microbes that can potentially enrich the STPs with harmful chemicals and will further assist in identifying means to ameliorate their toxic impacts.

The bacterial community have an increased potential to reduce nitrate to nitrite and its transportation outside the cell will be the main contributor to elevated TSNA levels during the storage of STPs (González et al. 2006). Earlier studies characterized nitrogen metabolism genes in American STPs and Sudanese Toombak by whole metagenome analysis and 16S rRNA gene metagenomics approach, respectively (Tyx et al. 2016; Rivera et al. 2020). The American dry snuff products demonstrated significant abundance of the nitrate reductase genes (*narGHJI*), nitrite reductase genes (*nirABC*) and nitrate/nitrite transporters genes (Tyx et al. 2016). In the current analysis, the samples of Pan masala, Zarda, and loose tobacco showed abundance of *narZJI* and *nirABDK*. Hence, the characterization of the bacterial community harboring a STP is crucial in deciphering their carcinogenic potential.

In the current study, the imputed metagenome of dry STPs like PM2, MK, D, G1, G2, and Z1 showed a high abundance of antibiotic resistance genes and several MFS transporter resistance genes. Previously, a study identified both Gram-positive and Gram-negative bacterial isolates obtained from waterpipe device hoses and characterized resistance to antibiotics like aztreonam, cefixime, norfloxacin, amoxicillin, clarithromycin, and enrofloxacin in some of the detected pathogens. The authors concluded that the hose of the waterpipe device can potentially be a good environment for the growth of bacterial pathogens, which can then be transmitted to their users (Masadeh et al. 2015). Another study conducted on STPs of American origin exhibited numerous antibiotic resistance genes and genes encoding for multidrug transporters and efflux pumps (Rivera et al. 2020). Hence, the dry and loose STPs can be the reservoir of antibiotic resistance genes and their consumption may lead to the spread of these genes to oral microbiota of SLT users, thereby making the treatment difficult.

LPS binding with the toll-like receptor (TLR receptor) can induce the production of cytokines -like IL-1β, IL-6, and TNF-α thereby generating an inflammatory response (Zhang and Ghosh 2000). Another study provided evidence that LPS-mediated chronic inflammation creates a favorable immunosuppressive microenvironment for tobacco-specific carcinogen-induced lung cancer progression (Liu et al. 2021). It has also been demonstrated that LPS exposure to co-cultures of oral squamous cell carcinoma (OSCC) cells with monocytes leads to STAT3 activation which has the potential to contribute to OSCC progression (Kurago et al. 2008). The genes related to LPS transport (ABC transporter complex, *lptBFG*) were dominant in the predicted metagenome of Pan masala, Z2, G2, and MK. Studies have also shown that bacterial flagellin contributes to the adhesion and invasion of host cells as well as acts as a potent immunomodulatory agent (Hajam et al. 2017). Clinical studies have demonstrated the role of bacterial products like LPS and flagellin in the development of colorectal cancer and hepatocellular carcinoma (Fedirko et al. 2017; Kong et al. 2016). Therefore, it is a possibility that constant exposure of the buccal lining of SLT users to LPS and flagellin can cause chronic inflammation leading to carcinogenesis.

Bacterial toxins (endotoxins and exotoxins) possess the potential to damage the host at the site of infection or distant from the site (Henkel et al. 2010). Chewing of SLT not only exposes the buccal mucosa but also its ingestion can lead to serious consequences. Bacterial toxins are inflammagens known to induce cytokine storm; the endotoxins trigger TLR4/MD2 expressed on the surface of monocytes, dendritic cells, and macrophages via CD14, whereas exotoxins induce the release of T-lymphocytes derived cytokines (Cavaillon 2018). Gram-positive and Gram-negative bacteria express hemolysins which are protein toxins that enable pore formation in biological membranes thereby lysing eukaryotic cells (Braun and Focareta 1991). The Pan masala products were observed to have a high prevalence of the hemolysin genes. Another bacterial gene product, pneumolysin (thio-activated cytolysin, K11031), is crucial in *Streptococcus pneumonia* mediated pathogenesis and has been shown to cause acute lung injury and cause necrotic and apoptotic cell death in the naso-pharyngeal passage (Nishimoto et al. 2020). In our study, the abundance of genera *Streptococcus*, as well as the pneumolysin gene, was observed to be high in the product G1. Further, the products PM3 and Z2 showed an abundance of neuraminidase (sialidase) gene which is expressed by mucosal pathogens like *Hemophilus influenzae, S. pneumoniae,* and *Pseudomonas aeruginosa* (Soong et al. 2006). Therefore, the presence of these bacterial toxins in SLT products is noteworthy, as they possess the potential to generate host tissue injury, especially in the oral and nasal epithelial layer.

The MicrobiomeAnalyst tool was used to draw correlations between the STP bacteriome with human-intrinsic factors. The genera *Streptococcus, Haemophilus, Moraxella* and *Pseudomonas* are typical members of the lung microbiota prevalent in COPD (Wang et al. 2016) whereas, the genera *Escherichia coli, Escherichia faecalis, F. nucleatum*, and *Streptococcus gallolyticus* inhabit the colon in high abundance during colorectal cancer (Alhinai et al. 2019). In this study, the bacteriome of dry STPs had correlated with increased risk of colorectal cancer, Type I diabetes and COPD in humans. Moreover, the abundance of the above-mentioned genera particularly in Pan masala products, one of the Gul samples (G1) and MK, can contribute to dysbiosis, thereby disposing the SLT users to the risk of disease development.

In this study, the core bacteriome of STPs correlated with that of mucin-degraders and indole producers. The mucus lining on the respiratory and gastrointestinal tract is composed of large glycoproteins, called mucins, which protects the underlying epithelial cells from mechanical abrasions as well as inhibit the entry of harmful substances, such as drugs, toxins and microbial pathogens. This mucus layer serves as the initiation surface for host-microbe interaction by providing attachment sites to bacteria. Mucin degrading bacterial utilize the mucus as a carbon source for growth thereby disturbing the protection of the host mucosal surfaces (Derrien et al. 2010). The genera *Ruminococcus* and *Bifidobacterium* which have been abundantly reported in the various STPs in the present study were previously shown to possess mucin degradation ability (Miller and Hoskins 1981). Additionally, several pathogenic bacteria, like *Vibrio vulnificus*, *H. influenzae*, *Shigella* strains, *Klebsiella planticola,* and *Proteus vulgaris,* are known indole producers and these genera have also been reported in the STPs analyzed in this study. Indole, a natural biomolecule produced by many Gram-positive and Gram-negative bacteria, has been shown to play diverse roles in bacterial spore formation, drug resistance, virulence, plasmid stability, and biofilm formation (Lee and Lee 2010).

Furthermore, the outcomes of this study have few limitations. Firstly, the study samples have been procured from a restricted geographical location and comprise brands popular in this region as well as local products manufactured in this area only. Therefore, the microbiological composition may vary in other products available across the country owing to varied climatic conditions which influence the pH and moisture of these products during processing and storage. Secondly, this study focuses on the bacteriome of these products, however, there are several evidence in the published literature that fungal species also thrive in SLTs which may add to the carcinogen load by producing mycotoxins and contributing to TSNA formation (Rivera et al. 2020; Saleem et al. 2018). All in all, the findings of the present study strongly suggests that the manufacturing of STPs should be coupled with thorough quality control protocols starting from harvesting to processing and storage and stringent measures should be taken to regulate the manufacturing practices of locally available loose SLT products.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00253-022-11979-y.

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Author Contribution SS wrote the first draft. MS analyzed the data. HS reviewed and edited the manuscript. MB designed the research and acquired the funding. All authors read and approved the manuscript.

Data Availability The datasets generated during the current study was submitted to the Short Read Archive of NCBI (BioProject accession number PRJNA787750) and is also available from the corresponding author on reasonable request.

Declarations

Conflict of Interest The authors declare that there is no financial or non-financial conflict of interest.

References

Agerholm J, Jensen N, Giese S, Jensen H (1997) A preliminary study on the pathogenicity of *Bacillus licheniformis* bacteria in immunodepressed mice. APMIS : Acta Pathologica, Microbiologica, Et Immunologica Scandinavica 105(1–6):48–54

Al-Hebshi NN, Alharbi FA, Mahri M, Chen TJG (2017) Differences in the bacteriome of smokeless tobacco products with different oral carcinogenicity: compositional and predicted functional analysis. Genes 8(4):106. https://doi.org/10.3390/genes8040106

Alhinai EA, Walton GE, Commane DM (2019) The role of the gut microbiota in colorectal cancer causation. Int J Mol Sci 20(21):5295. https://doi.org/10.3390/ijms20215295

Andrews S (2010) FastQC: a quality control tool for high throughput sequence data. Babraham Bioinformatics, Babraham Institute, Cambridge, United Kingdom. https://www.bioinformatics.babraham.ac.uk/projects/fastqc/

Braun V, Focareta T (1991) Pore-forming bacterial protein hemolysins (cytolysins). Crit Rev Microbiol 18(2):115–158. https://doi.org/10.3109/10408419109113511

Caporaso JG, Kuczynski J, Stombaugh J, Bittinger K, Bushman FD, Costello EK, Fierer N, Peña AG, Goodrich JK, Gordon JI, Huttley GA, Kelley ST, Knights D, Koenig JE, Ley RE, Lozupone CA, McDonald D, Muegge BD, Pirrung M, Reeder J, Sevinsky JR, Turnbaugh PJ, Walters WA, Widmann J, Yatsunenko T, Zaneveld J, Knight R (2010) QIIME allows analysis of high-throughput community sequencing data. Nat Methods 7(5):335–336. https://doi.org/10.1038/nmeth.f.303

Cavaillon J-M (2018) Exotoxins and endotoxins: Inducers of inflammatory cytokines. Toxicon 149:45–53. https://doi.org/10.1016/j.toxicon.2017.10.016

Chattopadhyay S, Malayil L, Mongodin EF, Sapkota AR (2021) A roadmap from unknowns to knowns: Advancing our understanding of the microbiomes of commercially available tobacco products. Appl Microbiol Biotechnol 105(7):2633–2645. https://doi.org/10.1007/s00253-021-11183-4

Chong J, Liu P, Zhou G, Xia J (2020) Using MicrobiomeAnalyst for comprehensive statistical, functional, and meta-analysis of microbiome data. Nat Protoc 15(3):799–821. https://doi.org/10.1038/s41596-019-0264-1

Cigarettes and other tobacco products act (COTPA) (2003) Restrictions on trade and commerce in, and production, supply and distribution of cigarettesand other tobacco products. The cigarettes and other tobacco products act. Gazette of India, India, Sect. 7(5). https://legislative.gov.in/sites/default/files/A2003-34.pdf

De Waal GM, De Villiers WJ, Forgan T, Roberts T, Pretorius E (2020) Colorectal cancer is associated with increased circulating lipopolysaccharide, inflammation and hypercoagulability. Sci Rep 10(1):1–20. https://doi.org/10.1038/s41598-020-65324-2

Derrien M, van Passel MW, van de Bovenkamp JH, Schipper RG, de Vos WM, Dekker J (2010) Mucin-bacterial interactions in the human oral cavity and digestive tract. Gut Microbes 1(4):254–268. https://doi.org/10.4161/gmic.1.4.12778

Dhariwal A, Chong J, Habib S, King IL, Agellon LB, Xia J (2017) MicrobiomeAnalyst: a web-based tool for comprehensive statistical, visual and meta-analysis of microbiome data. Nucleic Acids Res 45(W1):W180-w188. https://doi.org/10.1093/nar/gkx295

Ewels P, Magnusson M, Lundin S, Käller MJB (2016) MultiQC: summarize analysis results for multiple tools and samples in a single report. Bioinformatics 32(19):3047–3048. https://doi.org/10.1093/bioinformatics/btw354

Fedirko V, Tran HQ, Gewirtz AT, Stepien M, Trichopoulou A, Aleksandrova K, Olsen A, Tjønneland A, Overvad K, Carbonnel F (2017) Exposure to bacterial products lipopolysaccharide and flagellin and hepatocellular carcinoma: a nested case-control study. BMC Med 15(1):1–12. https://doi.org/10.1186/s12916-017-0830-8

Friedman J, Alm EJ (2012) Inferring correlation networks from genomic survey data. PLoS Comput Biol 8(9):e1002687. https://doi.org/10.1371/journal.pcbi.1002687

GATS-2 (2016–17) (2017) Global Adult Tobacco Survey-2 Factsheet India 2016–17. World Health Organization. https://ntcp.nhp.gov.in/assets/document/surveys-reports-publications/Global-Adult-Tobacco-Survey-Second-Round-India-2016-2017.pdf

Gonzalez PJ, Correia C, Moura I, Brondino CD, Moura JJG (2006) Bacterial nitrate reductases: molecular and biological aspects of nitrate reduction. J Inorg Biochem 100(5–6):1015–1023. https://doi.org/10.1016/j.jinorgbio.2005.11.024

Hajam IA, Dar PA, Shahnawaz I, Jaume JC, Lee JH (2017) Bacterial flagellin—a potent immunomodulatory agent. Exp Mol Med 49(9):e373–e373. https://doi.org/10.1038/emm.2017.172

Han J, Sanad YM, Deck J, Sutherland JB, Li Z, Walters MJ, Duran N, Holman MR, Foley SL (2016) Bacterial populations associated with smokeless tobacco products. Appl Environ Microbiol 82(20):6273–6283. https://doi.org/10.1128/aem.01612-16

Henkel JS, Baldwin MR, Barbieri JT (2010) Toxins from bacteria. In: Luch A (ed) Molecular, clinical and environmental toxicology, vol 2. Clinical toxicology. Birkhäuser Basel, Basel, pp 1–29

Kanehisa M, Goto SJNar (2000) KEGG: Kyoto encyclopedia of genes and genomes . Nucleic Acids Res 28(1):27–30 https://doi.org/10.1093/nar/28.1.27

Kaur J, Sharma A, Kumar A, Bhartiya D, Sinha DN, Kumari S, Gupta R, Mehrotra R, Singh H (2019) SLTChemDB: A database of chemical compounds present in smokeless tobacco products. Sci Rep 9(1):7142. https://doi.org/10.1038/s41598-019-43559-y

Kong SY, Tran HQ, Gewirtz AT, McKeown-Eyssen G, Fedirko V, Romieu I, Tjønneland A, Olsen A, Overvad K, Boutron-Ruault M-C (2016) Serum endotoxins and flagellin and risk of colorectal cancer in the European Prospective Investigation into Cancer and Nutrition (EPIC) Cohort. Cancer Epidemiol Biomarkers Prev 25(2):291–301. https://doi.org/10.1158/1055-9965.EPI-15-0798

Krueger F (2015) Trim galore: A wrapper tool around Cutadapt FastQC to consistently apply quality adapter trimming to FastQ files. vol 516. Babraham Bioinformatics, Cambridge, United Kingdom. https://www.bioinformatics.babraham.ac.uk/projects/trim_galore/

Kurago ZB, Lam-ubol A, Stetsenko A, De La Mater C, Chen Y, Dawson DV (2008) Lipopolysaccharide-squamous cell carcinoma-monocyte interactions induce cancer-supporting factors leading to rapid STAT3 activation. Head Neck Pathol 2(1):1–12. https://doi.org/10.1007/s12105-007-0038-x

La Rosa GRM, Gattuso G, Pedullà E, Rapisarda E, Nicolosi D, Salmeri M (2020) Association of oral dysbiosis with oral cancer development. Oncol Lett 19(4):3045–3058. https://doi.org/10.3892/ol.2020.11441

Langille MGI, Zaneveld J, Caporaso JG, McDonald D, Knights D, Reyes JA, Clemente JC, Burkepile DE, Vega Thurber RL, Knight R, Beiko RG, Huttenhower C (2013) Predictive functional profiling of microbial communities using 16S rRNA marker gene sequences. Nat Biotechnol 31(9):814–821. https://doi.org/10.1038/nbt.2676

Lee J-H, Lee J (2010) Indole as an intercellular signal in microbial communities. FEMS Microbiol Rev 34(4):426–444. https://doi.org/10.1111/j.1574-6976.2009.00204.x

Liu C-H, Chen Z, Chen K, Liao F-T, Chung C-E, Liu X, Lin Y-C, Keohavong P, Leikauf GD, Di YP (2021) Lipopolysaccharide-mediated chronic inflammation promotes tobacco carcinogen–induced lung cancer and determines the efficacy of immunotherapy. Cancer Res 81(1):144–157. https://doi.org/10.1158/0008-5472.CAN-20-1994

Masadeh MM, Hussein EI, Alzoubi KH, Khabour O, Shakhatreh MAK, Gharaibeh M (2015) Identification, characterization and antibiotic resistance of bacterial isolates obtained from waterpipe device hoses. Int J Environ Res Public Health 12(5):5108–5115. https://doi.org/10.3390/ijerph120505108

Mehra R, Mohanty V, Balappanavar AY, Kapoor S (2020) Bacterial contamination of packaged smokeless tobacco sold in India. Tob Prev Cessat 6:11. https://doi.org/10.18332/tpc/115064

Melkamu T, Qian X, Upadhyaya P, O'Sullivan MG, Kassie F (2013) Lipopolysaccharide enhances mouse lung tumorigenesis: a model for inflammation-driven lung cancer. Vet Pathol 50(5):895–902. https://doi.org/10.1177/0300985813476061

Miller RS, Hoskins LC (1981) Mucin degradation in human colon ecosystems: Fecal population densities of mucin-degrading bacteria estimated by a "most probable number" method. Gastroenterology 81(4):759–765

Monika S, Dineshkumar T, Priyadharini S, Niveditha T, Rajkumar K (2020) Smokeless tobacco products (STPs) harbour bacterial populations with potential for oral carcinogenicity. Asian Pac J Cancer Prev 21(3):815–824. https://doi.org/10.31557/APJCP.2020.21.3.815

Mukherjea A, Modayil MV, Tong EK (2015) Paan (pan) and paan (pan) masala should be considered tobacco products. Tob Control 24(e4):e280–e284. https://doi.org/10.1136/tobaccocontrol-2014-051700

Nasrin S, Chen G, Watson CJ, Lazarus P (2020) Comparison of tobacco-specific nitrosamine levels in smokeless tobacco products: High levels in products from Bangladesh. PLoS ONE 15(5):e0233111. https://doi.org/10.1371/journal.pone.0233111

Nethan ST, Kumar V, Sharma S, Hariprasad R, Mehrotra R (2018) Prevalence of Gul use, its predictors and association with oral potentially malignant disorders and oral cancer development in the users of Noida, India: A cross-sectional study. J Glob Oncol 4(Supplement 2):24. https://doi.org/10.1200/jgo.18.31700

Nishimoto AT, Rosch JW, Tuomanen EI (2020) Pneumolysin: pathogenesis and therapeutic target. Front Microbiol 11:1543. https://doi.org/10.3389/fmicb.2020.01543

Reynolds-Campbell G, Nicholson A, Thoms-Rodriguez CA (2017) Oral bacterial infections: Diagnosis and management. Dent Clin North Am 61(2):305–318. https://doi.org/10.1016/j.cden.2016.12.003

Rivera AJ, Tyx RE, Keong LM, Stanfill SB, Watson CH (2020) Microbial communities and gene contributions in smokeless tobacco products. Appl Microbiol Biotechnol 104(24):10613–10629. https://doi.org/10.1007/s00253-020-10999-w

Rumi F (2019) Nicotine found in seven pan masala brands Times of India. Patna, India

Sajid M, Srivastava S, Joshi L, Bharadwaj M (2021a) Impact of smokeless tobacco-associated bacteriome in oral carcinogenesis. Anaerobe 70:102400. https://doi.org/10.1016/j.anaerobe.2021

Sajid M, Srivastava S, Kumar A, Kumar A, Singh H, Bharadwaj M (2021b) Bacteriome of moist smokeless tobacco products consumed in India with emphasis on the predictive functional potential. Front Microbiol 12: 784841. https://doi.org/10.3389/fmicb.2021.784841

Saleem S, Naz SA, Shafique M, Jabeen N, Ahsan SW (2018) Fungal contamination in smokeless tobacco products traditionally consumed in Pakistan. J Pak Med Assoc 68(10):1471–1477

Shetty PJ, Hegde V (2015) Assessment of pathogenic micro-organisms in commercially available smokeless tobacco products. Avicenna J Med 5(2):54–55. https://doi.org/10.4103/2231-0770.154202

Singh H (2020) Smokeless tobacco (SLT) products. WHO FCTC Knowledge Hub on Smokeless Tobacco.https://extranet.who.int/fctcapps/fctcapps/fctc/kh/slt/news/smokeless-tobacco-slt-products

Song M-A, Marian C, Brasky TM, Reisinger S, Djordjevic M, Shields PG (2016) Chemical and toxicological characteristics of conventional and low-TSNA moist snuff tobacco products. Toxicol Lett 245:68–77. https://doi.org/10.1016/j.toxlet.2016.01.012

Soong G, Muir A, Gomez MI, Waks J, Reddy B, Planet P, Singh PK, Kanetko Y, Wolfgang MC, Hsiao Y-S (2006) Bacterial neuraminidase facilitates mucosal infection by participating in biofilm production. J Clin Invest 116(8):2297–2305. https://doi.org/10.1172/JCI27920

Stepanov I, Hecht SS, Ramakrishnan S, Gupta PC (2005) Tobacco-specific nitrosamines in smokeless tobacco products marketed in India. Int J Cancer 116(1):16–19. https://doi.org/10.1002/ijc.20966

Tricker A, Preussmann R (1988) The occurrence of *N*-nitro compounds in zarda tobacco. Cancer Lett 42(1–2):113–118. https://doi.org/10.1016/0304-3835(88)90247-9

Tyx RE, Stanfill SB, Keong LM, Rivera AJ, Satten GA, Watson CH (2016) Characterization of bacterial communities in selected smokeless tobacco products using 16S rDNA analysis. PLoS ONE 11(1):e0146939. https://doi.org/10.1371/journal.pone.0146939

Vishwakarma A, Verma D (2021) Microorganisms: crucial players of smokeless tobacco for several health attributes. Appl Microbiol Biotechnol 105(16):6123–6132. https://doi.org/10.1007/s00253-021-11460-2

Wang Z, Bafadhel M, Haldar K, Spivak A, Mayhew D, Miller BE, Tal-Singer R, Johnston SL, Ramsheh MY, Barer MR, Brightling CE, Brown JR (2016) Lung microbiome dynamics in COPD exacerbations. Eur Respir J 47(4):1082–1092. https://doi.org/10.1183/13993003.01406-2015

Zhang G, Ghosh S (2000) Molecular mechanisms of NF-κB activation induced by bacterial lipopolysaccharide through Toll-like receptors. J Endotoxin Res 6(6):453–457. https://doi.org/10.1179/096805100101532414